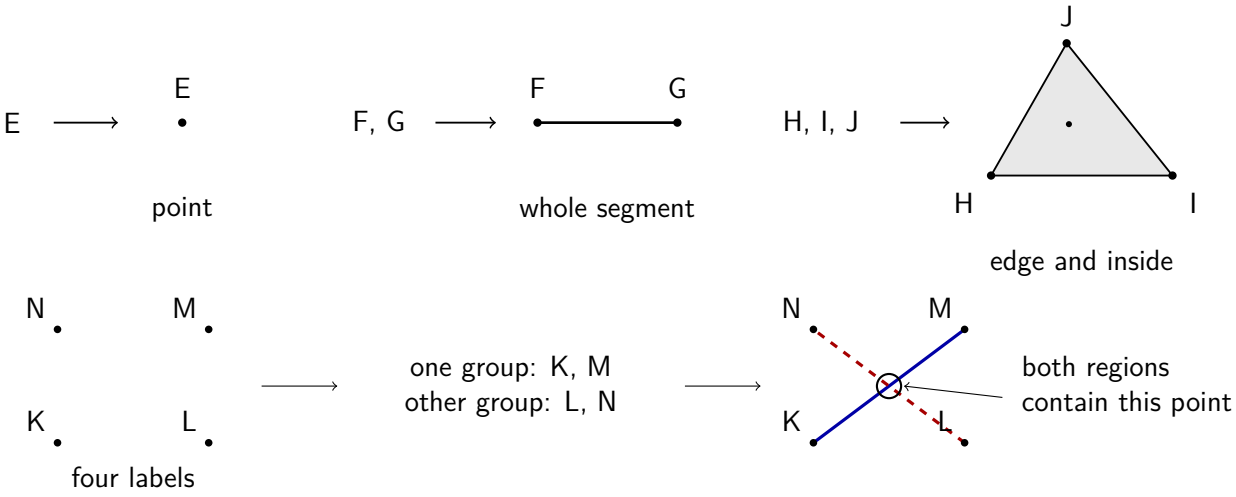
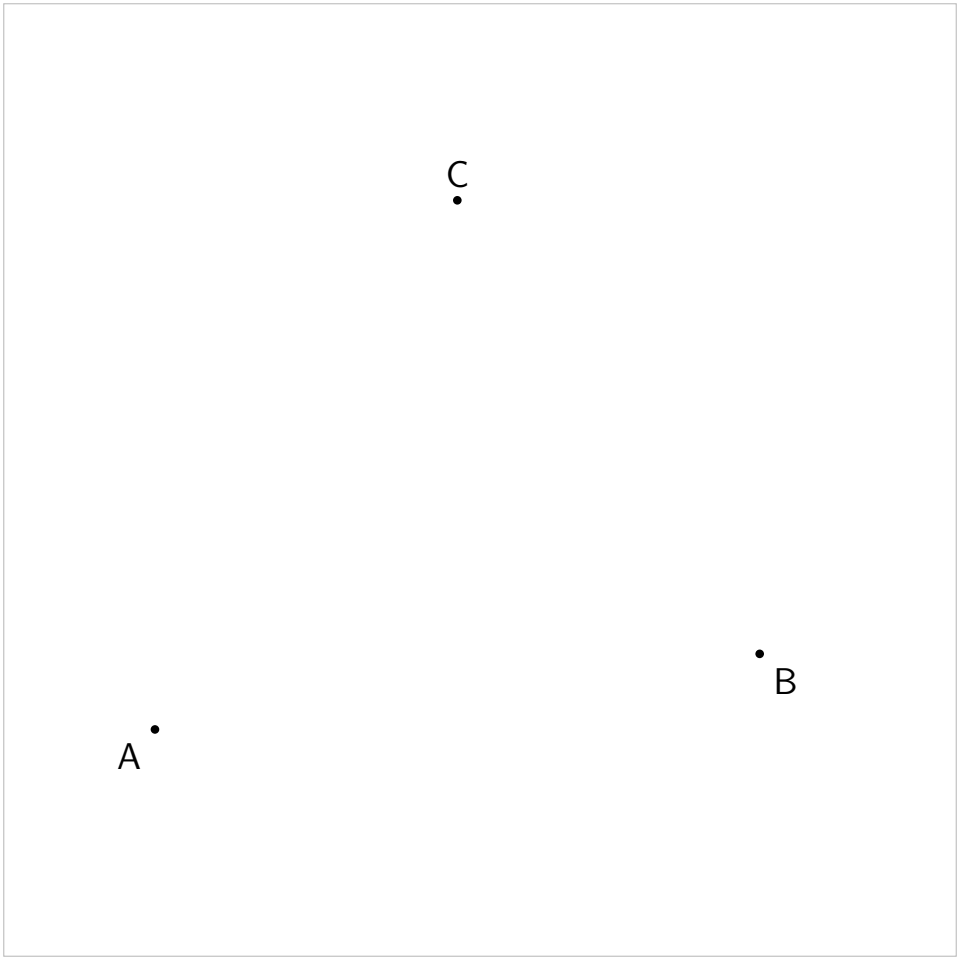
Use each label once in two groups, with at least one label in each. Draw each group's stretched-band region and lightly shade its inside. Use the centers of the marks. Sharing even one point counts. Swapping the two groups is the same split.



**Problem 5:** Can these three dots be split into two groups whose regions meet? Explain why your answer covers every split. Keep A and B fixed. Find every place in the box where C could go so that the regions meet.



**Problem 6:** Take turns placing A, B, C, and D in the box. Try to make an arrangement that the other player cannot split successfully. Can four dots always be split so their regions meet?